Transport in plants

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Xylem and phloem [1]

Which row shows the substances transported in xylem and in phloem?

	xylem	phloem
A	mineral ions and sucrose	water and amino acids
B	water and mineral ions	sucrose and amino acids
C	water and amino acids	sucrose and mineral ions
D	sucrose and amino acids	water and mineral ions

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2 · Part 1 Extended Xylem and phloem [1]

Which feature of xylem vessels allows them to form a long, continuous tube for transporting water?

A thick walls containing lignin

B many mitochondria in each cell

C sieve plates between neighbouring cells

D cells joined end to end with no cross walls

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3 · Part 1 Core Water uptake [1]

Water from the soil travels through a plant to a leaf.

Which sequence shows the pathway taken by the water?

A root hair cells → root cortex cells → xylem → mesophyll cells

B root cortex cells → root hair cells → xylem → mesophyll cells

C root hair cells → xylem → root cortex cells → mesophyll cells

D root hair cells → root cortex cells → phloem → mesophyll cells

4 · Part 1 Core Water uptake [1]

When young tomato plants are moved from small pots into a larger bed of soil, many of their root hairs are torn off.

Which statement describes the most likely effect on the plants in the first few days?

A More water is absorbed because the roots are now in a larger volume of soil.

B The xylem carries more water to the leaves than before.

C Less water and fewer mineral ions are absorbed because the surface area for uptake is smaller.

D More mineral ions are absorbed because the root cortex cells are now in direct contact with the soil.

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5 · Part 1 Core Transpiration [1]

Which statement describes how water leaves a leaf during transpiration?

A Liquid water diffuses out of the mesophyll cells and through the stomata.

B Liquid water is pumped out through the stomata by active transport.

C Water vapour passes directly from the xylem out through the waxy cuticle.

D Water evaporates from the surfaces of mesophyll cells into the air spaces and then diffuses out through the stomata as water vapour.

6 · Part 1 Core Transpiration [1]

The water uptake of a leafy shoot was measured with a potometer under four different conditions. The distance moved by the air bubble was used as a measure of the rate of transpiration. All other conditions were kept the same.

conditions	distance moved by air bubble in 10 minutes / mm
still air, 15 °C	12
still air, 25 °C	21
moving air from a fan, 15 °C	26
moving air from a fan, 25 °C	40

Which conclusion is supported by these results?

A Increasing the temperature and increasing the wind speed both increase the rate of transpiration.

B Increasing the temperature increases the rate of transpiration, but wind speed has no effect.

C Increasing the wind speed decreases the rate of transpiration.

D The rate of transpiration is highest in still air at 25 °C.

7 · Part 1 Extended Transpiration [1]

In a tall plant, a bubble of air forms inside one xylem vessel and completely fills a short section of it. Water stops moving up that vessel, although transpiration continues from the leaves.

Which statement explains why water stops moving up this vessel?

A The air bubble blocks the stomata, so transpiration stops in the whole plant.

B Mineral ions can no longer be actively transported into that vessel.

C The column of water is broken, so the transpiration pull can no longer draw water molecules up past the gap.

D The air bubble removes the lignin from the vessel walls, so the vessel collapses.

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8 · Part 1 Extended Transpiration [1]

Which change would decrease the rate of transpiration from a plant, and why?

A increasing the wind speed, because water vapour is removed from around the leaf more quickly

B increasing the humidity, because the water vapour concentration gradient between the air spaces and the air outside the leaf becomes smaller

C increasing the temperature, because water molecules gain more kinetic energy

D decreasing the humidity, because the air outside the leaf contains less water vapour

9 · Part 1 Extended Translocation [1]

Which statement describes translocation?

A the movement of sucrose and amino acids in phloem from parts that release them to parts that use or store them

B the movement of water and mineral ions in xylem from the roots to the leaves

C the movement of sucrose in xylem from parts that store it to parts that make it

D the loss of water vapour from the leaves through the stomata

10 · Part 1 Extended Translocation [1]

Potato plants store starch in underground tubers. In late summer the plant has many leaves and its tubers are growing. In the following spring, a stored tuber grows new shoots before these shoots have any leaves.

Which row describes the tuber at each time?

	tuber in late summer	tuber in spring
A	source	source
B	source	sink
C	sink	sink
D	sink	source

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11 · Part 1 Extended Xylem and phloem [6]

Xylem and phloem are the two transport tissues in flowering plants.

(a) State the two substances that are transported in phloem. [1]

(b) Xylem transports water and mineral ions. State one other function of xylem. [1]

(c) Explain how the structure of xylem vessels is related to their function. [4]

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12 · Part 1 Core Water uptake [6]

Some students investigated the pathway of water through the above-ground parts of a plant. They stood the cut end of a leafy celery stalk in a beaker of water containing a blue dye. Every 20 minutes they measured how far the blue colour had travelled up the stalk from the cut end.

time / min	distance travelled by blue colour / mm
0	0
20	32
40	61
60	90

(a) Suggest why the students added a dye to the water. [1]

(b) At the end of the investigation the students cut thin slices across the stalk. The blue colour was found only in small groups of cells. Name the tissue that was stained blue. [1]

(c) Calculate the mean rate at which the blue colour travelled up the stalk over the 60 minutes. Give your answer in mm per min. [2]

(d) The students repeated the investigation with an identical celery stalk that had all of its leaves removed. Predict how the results would be different, and give a reason for your prediction. [2]

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13 · Part 1 Core Water uptake [5]

Two bean seedlings, X and Y, of the same mass were grown in the same soil. The table shows the total surface area of the root hairs of each seedling.

seedling	total surface area of root hairs / mm^2
X	150
Y	600

(a) Predict which seedling, X or Y, takes up water more quickly. Give a reason for your answer. [2]

(b) Outline the pathway taken by water from the soil to the mesophyll cells of a leaf. [3]

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14 · Part 1 Core Transpiration [8]

Some students used a potometer to investigate the effect of wind speed on the rate of transpiration of a leafy shoot. They placed an electric fan in front of the shoot. The fan had four settings, and a higher setting produced faster-moving air. For each setting they measured the distance that an air bubble moved along the potometer tube in 5 minutes. The room was kept at 22 °C.

fan setting	distance moved by air bubble in 5 minutes / mm
0 (off)	14
1	31
2	47
3	55

(a) Calculate the rate of movement of the air bubble at fan setting 2. Give your answer in mm per min. [2]

(b) Describe the effect of wind speed on the rate of transpiration shown by these results. [2]

(c) The potometer measures the rate at which the shoot takes up water. Suggest why this is not exactly the same as the rate of transpiration. [1]

(d) The students predicted that increasing the temperature would also increase the rate of transpiration. Describe how they could use the same apparatus to test this prediction. [3]

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15 · Part 1 Extended Transpiration [6]

The table shows information about the lower surface of the leaves of two plant species, P and Q, grown in the same conditions. The leaves of both species have the same surface area.

species	number of stomata per mm^2	mean width of open stoma / µm
P	250	8
Q	90	3

(a) State where in a leaf water evaporates during transpiration. [1]

(b) Predict which species, P or Q, loses water vapour from its leaves more quickly. Explain your answer using the data in the table. [3]

(c) The mesophyll cells inside a leaf are separated by many interconnecting air spaces. Explain how this arrangement affects the rate of water vapour loss from the leaf. [2]

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16 · Part 1 Extended Transpiration [6]

In a tree, water moves from the roots up to the leaves through xylem vessels in the trunk.

(a) Explain the mechanism by which water moves upwards in the xylem. [4]

(b) Oak trees lose all of their leaves in autumn. Suggest why very little water moves up the xylem of an oak tree in winter. [2]

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17 · Part 1 Extended Transpiration [7]

Four identical potted plants were used to investigate transpiration. Each pot was sealed in a plastic bag so that water could only be lost from the leaves. Each plant was weighed, kept for 6 hours in a chamber under one set of conditions, and then weighed again. All other conditions, including light intensity and wind speed, were the same for every plant.

Humidity is given as relative humidity. A higher value means that the air contains more water vapour.

plant	temperature / °C	relative humidity / %	mass lost in 6 hours / g
1	20	40	9.6
2	20	80	3.0
3	30	40	15.0
4	30	80	5.4

(a) Calculate the percentage decrease in mass lost when the relative humidity was increased from 40% to 80% at 20 °C. [2]

(b) Explain the effect of increasing humidity on the rate of transpiration. [3]

(c) Explain why plant 3 lost more mass than plant 1. [2]

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18 · Part 1 Extended Transpiration [5]

On a hot, dry afternoon, the leaves and young stems of some lettuce plants in a vegetable garden droop. The soil is watered in the evening, and by the next morning the plants are upright again.

(a) State the term used for this drooping of a plant. [1]

(b) Explain how and why the drooping occurs. [4]

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19 · Part 1 Extended Translocation [6]

Sugar beet plants store sucrose in a large swollen root during their first year of growth. In their second year, they use this store to grow a tall flowering shoot early in the spring.

(a) Describe what is meant by a sink in a plant. [1]

(b) Name the tissue in which sucrose is transported from the leaves to the root. [1]

(c) Explain why the swollen root acts as a sink in the first year but as a source in the second year. [4]

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20 · Part 1 Extended Translocation [6]

In an experiment, a complete ring of bark was removed from the main stem of a young woody plant, about halfway up the stem. Removing the ring of bark removes the phloem but leaves the xylem undamaged.

After several weeks, the stem had swollen just above the ring. The leaves above the ring had not wilted.

(a) Explain why the stem became swollen just above the ring. [2]

(b) Explain why the leaves above the ring had not wilted. [2]

(c) Suggest why the roots of this plant would eventually die. [2]

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